EXERCISE 4

Hotel Management Application

1. Based on the text, identify the actors of the system.

The actors are:

- Hotel client
- Receptionist
- Cashier
- Housekeeping staff
- Floor manager
- Maintenance staff
- Client's bank server

2. Identify the actions that occur outside the system.

- Making a reservation by phone: the phone conversation takes place outside the application.
- Payment verification and validation: the client's bank server performs this externally.
- Physical hotel activities: the client arrives and leaves; housekeeping staff clean rooms; the floor manager inspects rooms; maintenance staff repair problems.
- Delivering additional services: meals or other services are physically provided by the hotel.
- SSL: SSL is a security mechanism for communication, not a separate use case.

3. If we wanted to decompose the system into boundaries, what would these boundaries be?

The system can be divided into the following five functional boundaries, as shown in the diagram:

- **Reservation**
- **Billing**
- **Stay**
- **Cleaning**
- **Maintenance**

The outer rectangle represents the Hotel Management Application. The client's bank server is an external system, outside this boundary.

4. Draw the use case diagram of the system with functional boundaries (modules).

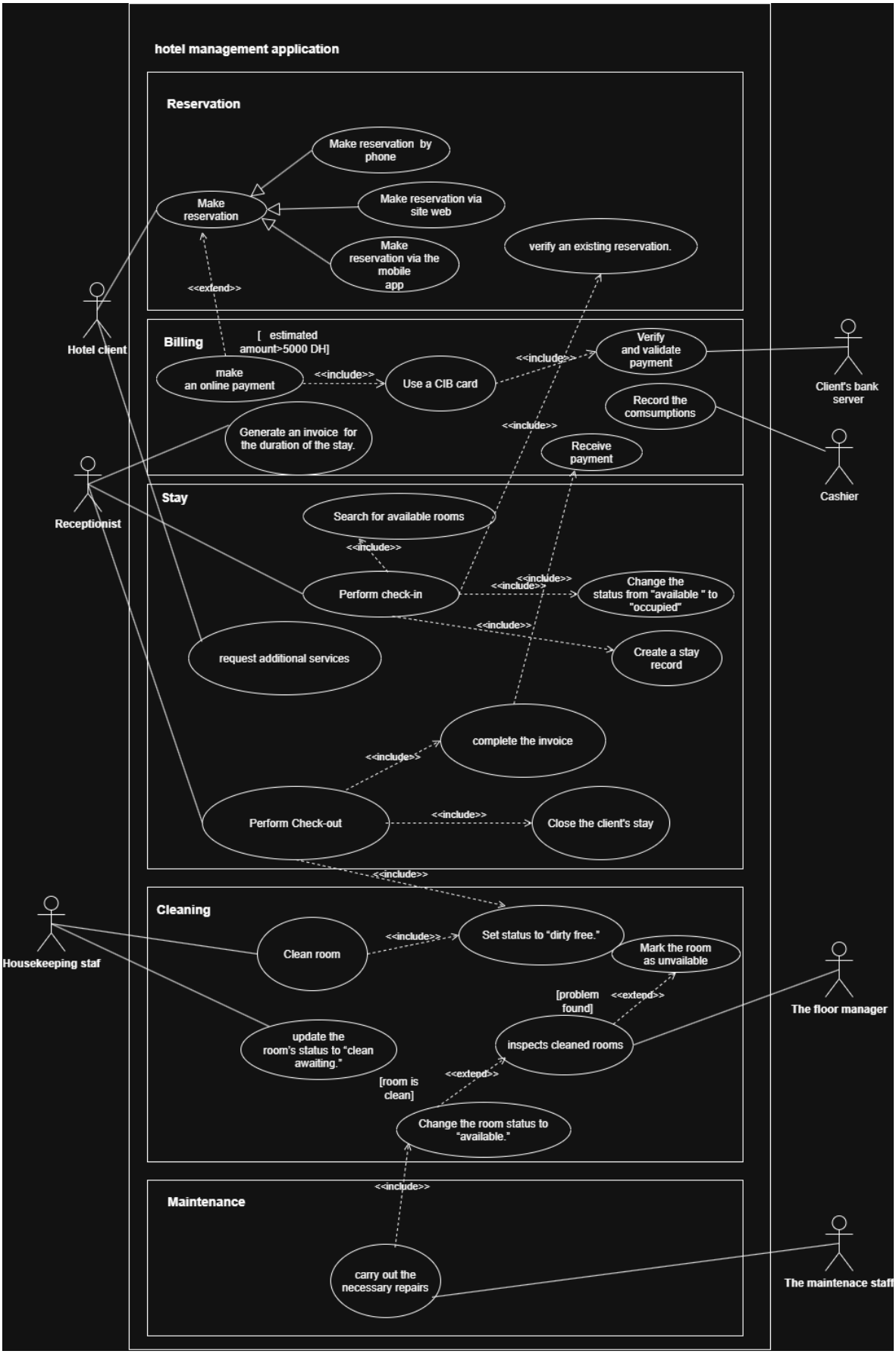


Figure 1. Hotel Management Application use case diagram.